Rajalakshmi Engineering College

FUNDAMENTALS OF DATA SCIENCE LAB

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EXPERIMENT 10

Understand array functions in Data Science

Experiment 10: Understand array functions in Data Science

Aim

To demonstrate array functions using NumPy.

Source Code

```
import numpy as np
array=np.random.randint(1,100,9)
array
np.sqrt(array)
array.ndim
new_array=array.reshape(3,3)
new_array
new_array.ndim
new_array.ravel()
newm=new_array.reshape(3,3)
newm
newm[2,1:3]
newm[1:2,1:3]
new_array[0:3,0:0]
new_array[0:2,0:1]
new_array[0:3,0:1]
new_array[1:3]
```

Output - Experiment 10

```
array([83, 25, 19, 47, 62, 15, 96, 39, 51])
array([9.11043358, 5. , 4.35889894, 6.8556546 , 7.87400787,
3.87298335, 9.79795897, 6.244998 , 7.14142843])
1
array([[83, 25, 19],
[47, 62, 15],
[96, 39, 51]])
2
array([83, 25, 19, 47, 62, 15, 96, 39, 51])
array([[83, 25, 19],
[47, 62, 15],
[96, 39, 51]])
array([39, 51])
array([[62, 15]])
array([], shape=(3, 0), dtype=int64)
array([[83],
[47]])
array([[83],
[47],
[96]])
array([[47, 62, 15],
[96, 39, 51]])
```

Result - Experiment 10

Array functions were demonstrated using the given examples.